Zeyu Xiao

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Education

- **Fudan University** — Shanghai, China
  Master's students in Computer Technology — Aug. 2023 – Present

- **Tianjin University** — Tianjin, China
  Bachelor of Computer Science and Technology — Aug. 2019 – Jul. 2023

Publications

[1] **Z Xiao***, Z Wu*, M Sun, Q Yan, Y Guo, Z Liang, L Zhang. PIG: Physically-based Multi-Material Interaction with 3D Gaussians. *The 15th ACM International Conference on Multimedia Retrieval*, 2025 (**Accepted**)

[2] **Z Xiao***, M Sun*, Y Cong, L Wang, D Kou, Z Wu, H Wang, D Yang, P Zhai, Z Wang, L Zhang. UniMGS: Unifying Mesh and 3D Gaussian Splatting with Single-Pass Rasterization and Proxy-Based Deformation. *AAAI*, 2026 (**Accepted**)

[3] Z Qian*, **Z Xiao***, Z Wu, D Yang, M Li, S Wang, S Wang, D Kou, L Zhang. Smcd: High realism motion style transfer via mamba-based diffusion. *The 33rd ACM International Conference on Multimedia*, 2025(**Accepted**)

[4] S Zhao*, **Z Xiao***, X Chen. KineST: A Kinematics-guided Spatiotemporal State Space Model for Human Motion Tracking from Sparse Signals. *AAAI*, 2026 (**Accepted**)

Research Experiences

- **Physics-driven Dynamic 3DGS Scene Generation**
  Dec. 2024 – May. 2025
  - We propose UniMGS, the first unified framework capable of rasterizing Mesh and 3DGS in a single-pass anti-aliased manner, incorporating a novel binding strategy for 3DGS deformation. Our work establishes a new paradigm for 3D scene generation and editing, while also unlocking novel possibilities for embodied AI, virtual reality, and interactive gaming.
  - The research[2] was accepted by AAAI 2026.

  May. 2024 – Aug. 2024
  - We propose the PIG framework, which innovatively integrates object-level segmentation of 3DGS with physics-based simulation, effectively addressing three key challenges in multi-material interactive scenes: inaccurate segmentation, deformation artifacts, and rendering inconsistencies. Our approach surpasses existing methods in both physical realism and visual fidelity, offering a novel solution for high-quality dynamic scene generation based on Material Point Method (MPM) simulation.
  - The research[1] was accepted by the 15th ACM International Conference on Multimedia Retrieval.

- **Virtual Digital Humans**
  Jan. 2025 - Apr. 2025
  - We propose UMSD, a unified Mamba-based framework for motion style transfer. Our single-stream architecture enables effective content-style feature interaction, solving the information loss and alignment bias in traditional two-stream approaches. Experiments show state-of-the-art performance in both short and long sequence tasks, benefiting digital human animation and film production.
  - The research[3] was accepted by the 33rd ACM International Conference on Multimedia.

  May. 2025 - Aug. 2025
  - We propose KineST, a novel kinematics-guided state space model for full-body motion tracking from sparse signals. Our framework integrates kinematics-guided bidirectional scanning and mixed spatiotemporal representation learning, solving the efficiency bottlenecks and decoupled modeling limitations of existing reconstruction methods.
  - The research[4] was accepted by AAAI 2026.